

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Viva Based Questions	Assessment:	Assessment Tool 3– Viva Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO Statement:	Demonstrate conceptual understanding of statistical inference, estimation, and hypothesis testing through oral explanation and reasoning. (BL4)		
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Section / Batch:	SLOT-D	Date:	20-08-2026

Code of Conduct

I SYED MUHAFIZ A(192472282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

A. Introduction to Statistical Inference

1. What is statistical inference?

It's the process of drawing conclusions about a population based on a sample of data. Instead of studying every single data point, I use the information in a sample to make an educated, quantified guess about the bigger picture.

2. Why is statistical inference important in data science?

Because in almost every real project I've worked on, I never have access to the full population — collecting every single data point is too expensive, too slow, or just impossible. Inference lets me make reliable predictions and business decisions from a manageable subset of data.

3. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics just summarizes and describes the data I already have in hand — things like the mean, median, or a bar chart. Inferential statistics goes a step further and uses that sample data to draw conclusions or make predictions about a larger population I haven't fully observed.

4. What is a population in statistics?

The population is the entire group I'm actually interested in studying — every single member, not just a subset. For example, all customers of a company, or every API request the service will ever receive.

5. What is a sample?

A sample is the smaller subset I actually collect and analyze from that population — ideally selected in a way (like random sampling) that represents the population well.

6. Why do data scientists usually work with samples instead of complete populations?

Because collecting data on an entire population is usually too costly, too slow, or literally impossible — some populations are even infinite or ongoing, like all future transactions a platform will ever process. Sampling gives me a practical, faster way to still get trustworthy insights.

7. What is a population parameter?

It's a fixed numerical value that describes some characteristic of the entire population — like the true population mean μ or the true proportion p . In practice I almost never know it exactly; that's precisely what I'm trying to estimate.

8. What is a sample statistic?

It's a numerical value I calculate directly from my sample data — like the sample mean $\bar{x}$ or sample proportion $\hat{p}$. I use it as my estimate of the corresponding population parameter.

9. Give a real-world example where statistical inference is used in industry.

A good one from my own work: estimating an e-commerce platform's true conversion rate from a sample of 600 visitors, then building a confidence interval around it so the business knows the plausible range for the real conversion rate across all visitors.

10. What are the two major activities involved in statistical inference?

Estimation — which includes both point estimation and interval estimation — and hypothesis testing.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

It's the school of thought where probability is defined as the long-run frequency of an event if I repeated the same experiment infinitely many times. The population parameter is treated as a fixed, unknown constant, and I use sample data to estimate it.

12. What is the basic idea behind frequentist statistics?

That if I repeated the same sampling process over and over, my estimates and confidence intervals would behave in a predictable, consistent way in the long run — and that long-run behavior is what justifies trusting the result from any single sample.

13. How does the frequentist approach differ from the Bayesian approach?

Frequentist treats the parameter as fixed and unknown, with probability describing the long-run frequency of the data given that fixed parameter. Bayesian treats the parameter itself as a random variable with a probability distribution — a “belief” — that gets updated as new data comes in, using Bayes' theorem.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The parameter is fixed — it just happens to be unknown to me. The data is what's random and varies from sample to sample.

15. What does long-run frequency mean in frequentist statistics?

It means if I repeat the same random experiment or sampling process over and over, the proportion of times something happens will converge to the true underlying probability.

16. Why is random sampling important in the frequentist approach?

Because it ensures every member of the population has a fair, known chance of being selected, which avoids bias and makes the sample properly represent the population — that's really the foundation the whole frequentist framework relies on.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No. In the frequentist view the parameter is a single, fixed true value — it doesn't have a probability distribution of its own. What's random is the sample I draw and the statistic I compute from it, not the parameter itself.

18. Give an example of a frequentist problem in a data science application.

Estimating the click-through rate of an ad campaign using a confidence interval built from a random sample of impressions, then running a hypothesis test to decide whether a new ad design performs significantly better than the old one.

C. Point Estimation

19. What is a point estimate?

A single value calculated from sample data that serves as my best guess for an unknown population parameter — for example, using the sample mean as my estimate of the population mean.

20. What is an estimator?

It's the general rule or formula I use to calculate an estimate from sample data — like the formula $\sum x/n$ for the sample mean, before I've even plugged in numbers.

21. What is the difference between an estimator and an estimate?

The estimator is the general formula or method; the estimate is the specific numeric value I actually get once I plug real sample data into that estimator.

22. What is the point estimator for the population mean?

The sample mean, $\bar{x} = \sum x / n$.

23. How do you estimate a population proportion from a sample?

I take the number of successes or occurrences in my sample and divide it by the total sample size: $\hat{p} = x / n$.

24. What is the point estimate of population variance?

The sample variance, $s^2 = \sum (x - \bar{x})^2 / (n - 1)$ — dividing by $n - 1$ instead of n is what makes it an unbiased estimator.

25. Why is the sample mean commonly used to estimate the population mean?

Because it's unbiased — its expected value equals the true population mean — it's simple to compute, and thanks to the Central Limit Theorem it behaves nicely, becoming approximately normal even when the underlying data isn't, as long as the sample is reasonably large.

26. What properties should a good estimator have?

It should be unbiased, have low variance (efficient), be consistent — meaning it gets closer to the true value as sample size grows — and ideally be sufficient, using all the relevant information in the sample.

27. What is an unbiased estimator?

One whose expected value, averaged over many repeated samples, equals the true population parameter — it doesn't systematically over- or under-estimate the truth.

28. What is meant by the bias of an estimator?

The difference between the expected value of the estimator and the true parameter value — it tells me whether the estimator systematically leans high or low on average.

29. What is the difference between bias and variance of an estimator?

Bias is about accuracy — whether the estimator is centered on the true value on average. Variance is about precision — how much the estimate would bounce around from sample to sample. Ideally I want both to be low, but in practice there's often a trade-off between the two.

30. Why can two different samples produce different point estimates for the same population parameter?

Because each sample is just a random subset of the population, so purely by chance it'll contain slightly different values — that's natural sampling variability, not necessarily a mistake in either sample.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Because it's based on just one particular sample, and a different random sample from the same population would almost always give a slightly different value — so there's always some inherent sampling variability baked into any single estimate.

32. What is the sampling distribution of an estimator?

It's the probability distribution I'd get if I repeatedly drew samples of the same size from the population and computed the estimator every time — it shows how the estimator varies across all possible samples.

33. What is the standard error?

It's the standard deviation of the sampling distribution of a statistic — basically, it measures how much a sample statistic like the sample mean is expected to vary from one sample to another.

34. How is standard error different from standard deviation?

Standard deviation measures the spread of individual data points around the mean within one dataset. Standard error measures the spread or uncertainty of a sample statistic itself — like the sample mean — across repeated samples; for the mean, $SE = \sigma/\sqrt{n}$.

35. How does increasing the sample size affect the standard error of the sample mean?

It decreases the standard error, since $SE = \sigma/\sqrt{n}$ — a larger sample pulls my estimate closer to the true value more consistently.

36. Why does variability in an estimate decrease when the sample size increases?

Because with more data points, unusual or extreme values get averaged out more effectively, so the sample statistic becomes more stable and less sensitive to any one odd observation.

37. What is the relationship between standard error and confidence intervals?

The width of a confidence interval is built directly from the standard error — $CI = \text{point estimate} \pm (\text{critical value} \times SE)$ — so a smaller standard error gives me a narrower, more precise interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

No, not necessarily. Sample means naturally vary from sample to sample just because of random sampling — that's expected variability, not an error, unless there's an actual problem like a biased sampling method or a data collection mistake behind one of them.

E. Confidence Intervals

39. What is a confidence interval?

A range of values, built around a point estimate, that's likely to contain the true population parameter at a certain confidence level — like 95%.

40. Why is a confidence interval more informative than a point estimate?

Because a point estimate alone gives no sense of how reliable or precise it is. A confidence interval shows the plausible range and directly communicates the uncertainty involved, giving a much fuller picture than a single number.

41. What does a 95% confidence level mean in the frequentist approach?

If I repeated the sampling process many times and built a confidence interval each time, about 95% of those intervals would contain the true population parameter. It's not saying there's a 95% probability the true value falls in this one specific interval I've already calculated.

42. What is the difference between a confidence level and a confidence interval?

The confidence level (like 95%) is the long-run success rate of the method used to generate intervals. The confidence interval is the actual numeric range — like (233, 247) — computed from one specific sample.

43. What factors determine the width of a confidence interval?

The sample size, the variability in the data (standard deviation), and the chosen confidence level — a bigger sample or lower variability narrows the interval, while a higher confidence level widens it.

44. How does increasing the sample size affect the width of a confidence interval?

It narrows the confidence interval, because the standard error decreases as sample size increases, and the interval's width is directly proportional to the standard error.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

The interval gets wider. To be more confident of actually capturing the true value, I need to cast a wider net — which means using a larger critical value (2.576 instead of 1.96).

46. What is the difference between a z-confidence interval and a t-confidence interval?

A z-interval assumes I know the population standard deviation, or that the sample is large enough that s is a good stand-in for it, and it uses the normal distribution. A t-interval is used when the population standard deviation is unknown and the sample is small, using the t-distribution, which has fatter tails to account for that extra uncertainty.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

When the population standard deviation is unknown and I'm working with a small sample — typically under about 30 observations. The t-distribution's fatter tails properly account for the added uncertainty of having to estimate σ using the sample standard deviation s .

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

A formal statistical method for deciding, using sample data, whether there's enough evidence to reject a claim — the null hypothesis — about a population parameter.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

H_0 , the null hypothesis, is the default assumption or status-quo claim I'm testing — usually stating there's no effect or no difference. H_1 , the alternative hypothesis, is what I'd conclude if there's enough evidence against H_0 — that there really is an effect or a difference.

50. How can a confidence interval be used to make a decision about a hypothesis?

If the hypothesized value under H_0 falls outside the confidence interval, I reject H_0 . If it falls inside the interval, I fail to reject H_0 . This gives exactly the same conclusion as running a formal significance test at the matching alpha level — it's just a more visual, intuitive way to get there.